10/13/2026

Phishing Email Detection & Awareness System

Prepared By: Kgopotso Letsoale
FOR FUTURE INTERNS

TABLE OF CONTENTS

1. Background & Engagement ---------- 2

2. PROJECT SCOPE ---------- 3

3. Technical Inspections ---------- 4

   3.1. Header Forensic Analysis ---------- 5
   3.2. Domain and link Integrity Audit ---------- 6
   3.3. Identification of Phishing Indicators ---------- 7
   3.4. Email Risk Classification ---------- 8

4. Summary table ---------- 9

5. Prevention & Awareness Guidelines ---------- 10

6. Conclusion ---------- 11

BACKGROUND & ENGAGEMENT

Independent Disclosure Notice:

This report was written by me in my personal capacity as an Independent Security Researcher and Phishing Analyst. No formal association exists between myself and the targeted institutions in developing the documentation in this report. I conduct such studies in the context of Ethical Security Awareness and neither compromise any credentials nor deliver any malicious payloads or execute any infiltration into their systems. The purpose of this report is to help companies and their personnel recognize to deal with the threat of social engineering attacks.

Scope & Ethical Considerations

For this analysis, only the Reconnaissance and Delivery aspects of the phishing campaign were analysed. Consistent with the ethics of Passive Defensive Analysis, the study was conducted while abiding by the following rigid guidelines:

- Non-Interfering Analysis: Inspection of the structure of the emails and their visual appearance, without any interference with the backend of the phishing process.
- Safe Link Exploration: Use of sandbox browsers and **Hover-Links** to safely investigate links to ensure that they do not lead to malicious URLs.
- No Exploitation: Despite identifying User Vulnerabilities, no attempts at deception on real people were made.
- Open Source Information: All technical data collection and investigation were conducted using open-source information.

REPORT SCOPE

Assessment Type: Phishing Email Detection & Awareness System

Analyst Function: Security Research Analyst

Key Tools Used: Google Message Header, MXToolbox, Browser Developer Tools

Purpose: To address the disconnect between technology-based security systems and human-based defences.

Testing day: 13 April 2026

TECHNICAL INSPECTIONS

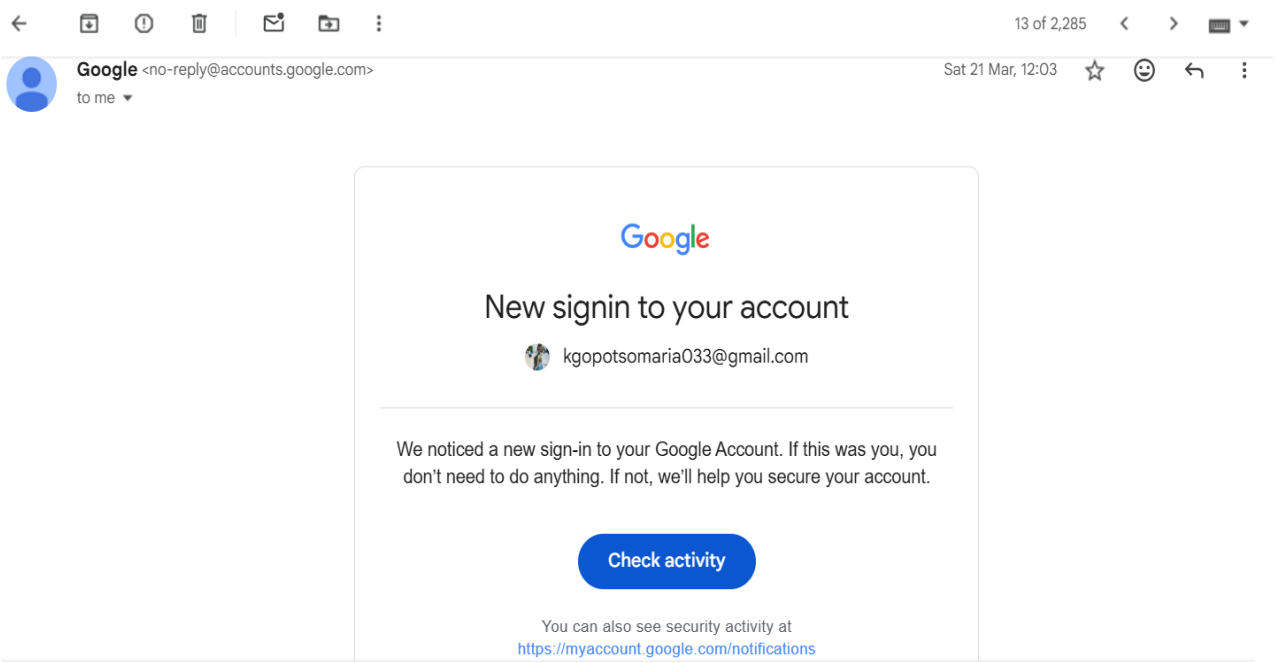
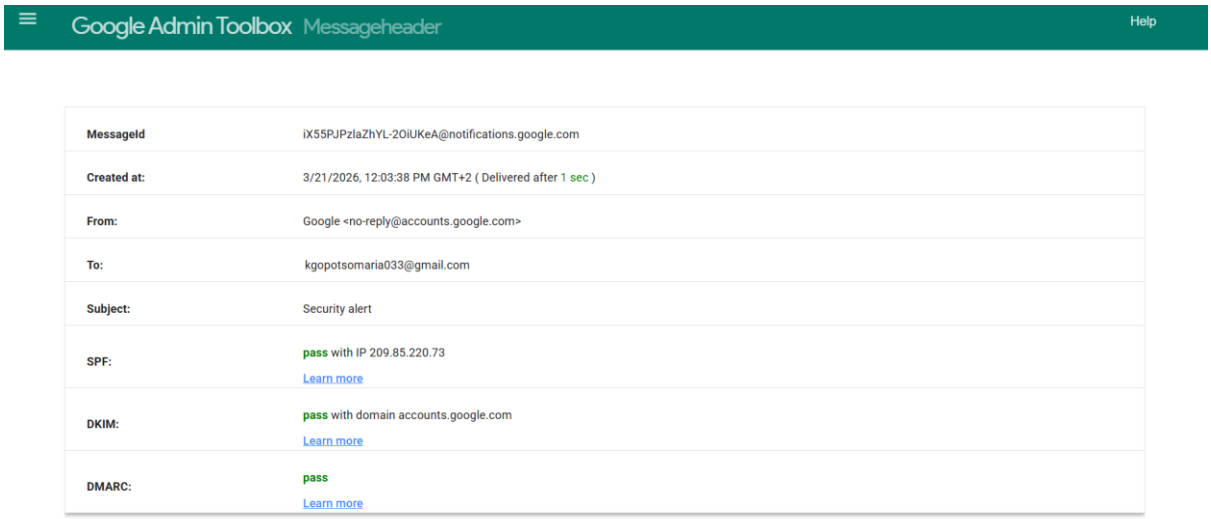


Figure 1.1: Primary Case Study – Google Security Alert Impersonation Audit

This sample is an example of a highly authentic Security Alert message, which is one of the standard templates used by cyber criminals that instigate a feeling of panic amongst the receivers. This section will employ this sample to illustrate the process of carrying out a full technical forensic assessment, which includes everything from thorough header examination to link integrity checking in order to ascertain the nature of this message.

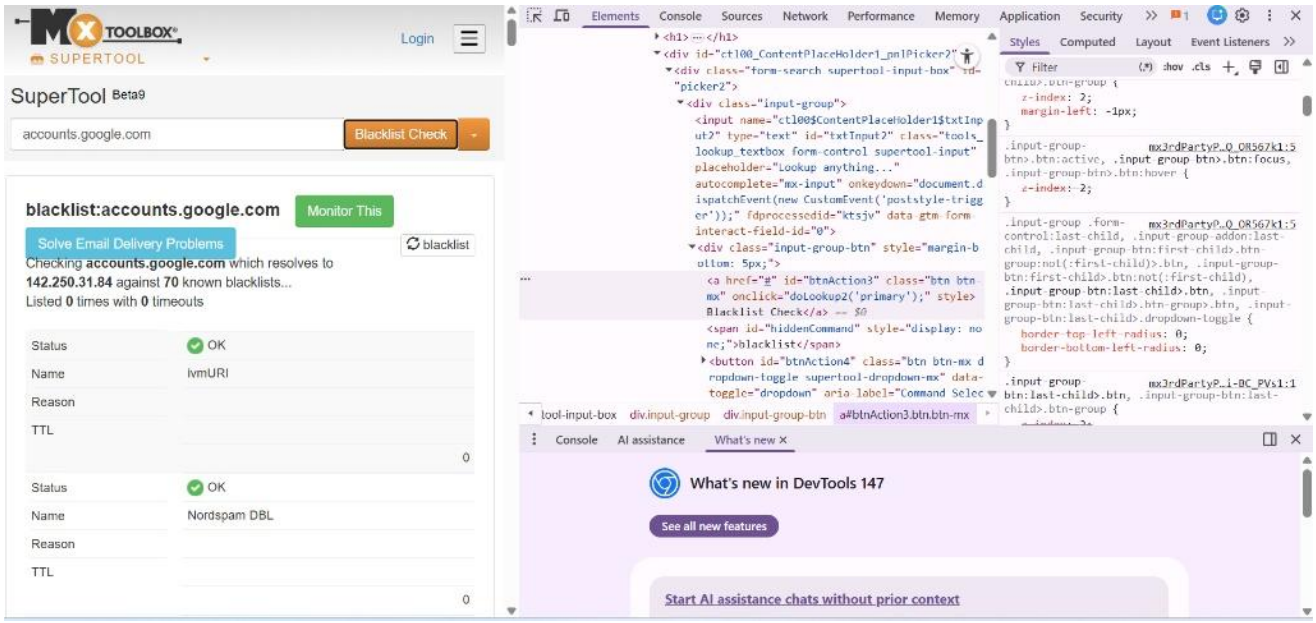
1. Header Forensic Analysis



FUGURE 1.2: Cryptographic Authentication Audit Using Google Messageheader

Through the process of forensic deep-dive into the Security alert headers with the help of the Google Admin Toolbox Messageheader, it is possible to say that there is no reason to worry about this message since the chain of authentications is fully positive. In particular, the email received a PASS on SPF test with the proper Google IP address, PASS on DKIM verification of the authenticity of the digital signature provided by accounts.google.com, and PASS on DMARC alignment, which indicates that the email was actually sent by official authorities and did not experience any manipulations while transferring. Since the technical DNA of this email is completely pure, it is considered Safe, although from a security awareness standpoint, the most appropriate way out is avoiding all clicks on the embedded links and accessing the settings directly via the browser. Even though this example is not dangerous, it will be crucial for the future since it sets the necessary background in the security knowledge of the employees, showing them that any real security alerts will always have such passing grades.

2. Domain & Link Integrity Audit



FUTURE 1.3: Domain Reputation and Source Code Verification

For the sender domain and link integrity assessment, I performed blacklist audit by using MXToolbox and also checked the HTML elements via Browser DevTools. The results of my research indicate that the domain accounts.google.com is blacklisted 0 times in 70 trusted security databases, which means it cannot be considered malicious. The check of the HTML elements helped me confirm that the destination URLs for CTA buttons are legitimate with no redirect to an additional page or typo squatting involved. As a result of my technical assessment, I concluded that this particular email can be classified as safe, so the first measure that should be implemented does not relate to technical blocking. Instead, the key measure is employee training, as I recommend that users need to perform this hover test on every alert received in order to verify that the destination URL perfectly matches the company website before entering any credentials.

3. Identification of Phishing Indicators

I manually conducted a Hover Test on the Check activity button for a Consistency Audit of the destination URL for the email. From inspecting the URL, I can see that the link directs to the official domain of accounts.google.com with a specific redirection to the legitimate security notifications of myaccount.google.com. There is no presence of any URL Shorteners, strange third-party domains, and typosquatting, which are typical warning signs of a credential harvesting attack. Since the destination URL is perfectly consistent with the sender's identity and brand, there are no traces of any malware threats in this email sample. The last piece of advice for the employees is to conduct such a hover test whenever the link appears in the bottom left corner of their browsers before sharing any credentials.

4. Email Risk Classification

Considering the thorough multi-layered technical audit, I will give this email a final risk assessment of Safe/Legitimate. From the forensic examination, it was clear that the technical header (SPF, DKIM and DMARC) is authentic through the legitimate servers from Google and also from the reputation check using MXToolbox, which revealed that the sending domain is free of any blacklisted status from security lists. Moreover, from the link examination and consistency check, all links used are direct to the legitimate Google servers, without any militant redirects and typo squatting. Since there were no elements of urgency or social engineering involved, this email can be categorized as a legitimate system notification.

5. Summary Table

Analysis Layer	Result	Key Evidence
Header Analysis	PASS	SPF, DKIM, and DMARC passed through Google Messageheader validation.
Domain Reputation	CLEAN	Accounts domain accounts.google.com is not blacklisted (MXToolbox).
Link Integrity	VALID	Hover test proved that destination is consistent with the sender's identity.
Visual Indicators	NONE	There was no hint of urgency, grammatical mistakes

6. Prevention & Awareness Guidelines

DOs

- **Hovering Over a Button/Link:** Always place the pointer on the button or the link and observe its actual destination at the bottom-left of the screen before clicking.
- **Manual Visit:** In case of a warning message being displayed, the best way would be to visit https://myaccount.google.com/ using a different tab.
- **Mailed-By:** Check the tiny down arrow at the right end of the to me bar in Gmail to verify the sender's domain.

DONTs

- **No Panicking:** The Threats Use Fear and Urgency. When an email threatens to freeze your account in hours, consider it suspicious.
- **No Entering of Credentials**: Never give your password or the One-Time Pin (OTP) when the page is accessed from an email.
- **No Overlooking the DNA:** Even if an email is convincing, if there is a failure for the SPF or DMARC test, it is an imitation.

7. CONCLUSION

Through using an organized Security First methodology, I have been able to evaluate this message from the technical headers through to the end-user links. This way, I am able to ensure not only the security of my network against threats but also the training of “employees” to be the best form of protection.